

Chinmay Singh

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EDUCATION

University of North Carolina at Chapel Hill, Chapel Hill, NC & **North Carolina State University**, Raleigh, NC May 2026
Candidate for Bachelor of Science

Major or Concentration: Biomedical Engineering, Concentrating in Medical Microdevices & Pharmacoengineering

Relevant Coursework: XR Engineering; Electroceuticals; Genetic Engineering; Biomaterials; Bioinstrumentation

Honors & Awards: NC State VenturePack Grand Prize & Audience Choice Winner for KAIRS; Global Runner-Up at 2024

Medtronic/BMES Student Design Competition for KAIRS; Gold Medal at iGEM Grand Jamboree 2025

Activities: Teaching Assistant for BMME 398: Junior Design and Manufacturing (Extended Reality and Cell Culture Modules);

Peer Mentor for BIOL 103: How Cells Function; Co-Host and Co-Founder of BioCast Podcast; EMT-Basic

North Carolina School of Science and Mathematics, Durham, North Carolina

May 2022

High School Diploma

RELEVANT EXPERIENCE

KAIRS, LLC | *Co-Founder*, Chapel Hill, North Carolina

August 2024 – Present

- KAIRS: Knee AI Rehab Sleeve: prototyped and validated solution for patients undergoing rehabilitation post-knee surgery, used 2 Inertial Measurement Units for angle tracking, developed TensorFlow model using CNN, achieving RMSE of ~0.81
- Raised \$13,500 in non-dilutive, equity-free funding across NC State VenturePack and WolfDen Challenges, placing 1st and 2nd, respectively

Sode Lab | *Undergraduate Researcher*, Chapel Hill, North Carolina

January 2025 – Present

- Researching single-chain variable fragment (scFv) antibody development by designing 6 candidates targeting NPY, narrowing to 4 targets for downstream validation
- Used AlphaFold 3 to model 12 proteins; evaluated confidence metrics (pLDDT/PAE), prioritized 4 for development

Stein & Zylka Labs | *Undergraduate Researcher*, Chapel Hill, North Carolina

June 2022 – December 2024

- Compared brain organoid differentiation protocols, increasing efficiency across the lab for cell culture by 70%
- Studying the effects of spinal macrophages on resolving nociceptive hypersensitivity utilizing immunohistochemistry techniques and flow cytometry in 3 batches in 96-well plates, quantifying 288 cell populations across conditions
- Built a Fiji/ImageJ quantification workflow to count cells across 3000+ images/sections, reducing analysis time from 5 minutes to 30 seconds per sample and improving quantification consistency.

LEADERSHIP

UNC iGEM | *Wet Lab Team Lead*, Chapel Hill, North Carolina

January 2025 – Present

- Led wet-lab sub-team of 6, running bi-weekly meetings and coordinating 5+ experiments/week; wrote protocols, task assignments, and data review to hit 100% of project milestones by 10/27/25.
- Presented poster at iGEM Grand Jamboree (Paris, France), poster accepted at CMBE 2026 (San Juan, PR)

Engineering World Health @ UNC | *President*, Chapel Hill, North Carolina

August 2023 – Present

- Organized and instructed kit-building events of 20+ to build optical heart rate monitor kits, soldering skills, and other electronic components, organized UNC's first Engineer-A-Thon, managing corporate-sponsored budget of \$2,000+

SKILLS

Technical: Unreal Engine 5, Unity 6, Python, Soldering, PCB Design, OnShape CAD, Adobe Suite, 3-D Printing, Laser Cutting

Languages: Hindi (fluent)

Interests: Martial arts (black belt in Taekwondo), healthcare, hardware/electronics, reading